Two-parameter continuation in XPPAUT-Web

Following a Hopf bifurcation of the Morris–Lecar model in two parameters with AUTO, in the browser

XPPAUT-Web runs the real XPPAUT (with AUTO) in a web browser: nothing to install, and it works on phones and tablets too. This guide traces how the Hopf bifurcation of the Morris–Lecar neuron moves when two parameters change: the applied current `iapp` and the rate `phi`.

Open the model: https://zetaone.github.io/XPPAUT-Web/?ode=ode/lecar.ode

A. Settle onto a steady state

1. Press `I` `G` (Initialconds → Go), then `I` `L` twice (Initialconds → Last), so the system sits at rest.

B. One-parameter diagram in `iapp`

2. Press `F` `A` (File → Auto) to open the AUTO window.
3. Click **Run → Steady state**. The diagram shows labels 1–5: two folds (LP, labels 2 and 3) and a **Hopf point (HB) at label 4**, at `iapp ≈ 0.204`.

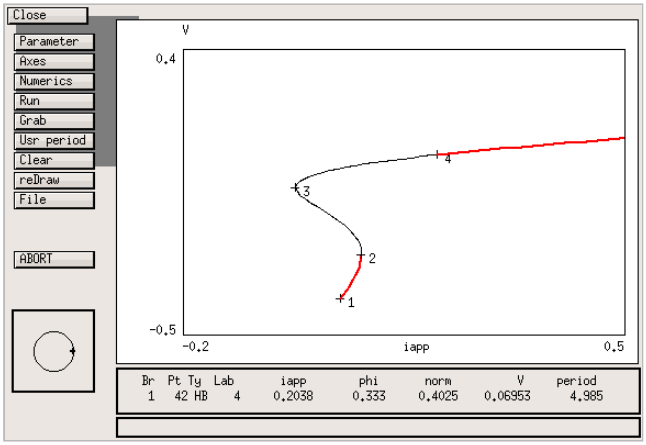


Figure 1. Steady states against `iapp`. Red: stable; black: unstable. The bottom line shows the grabbed point: Lab 4, type HB.

C. Continue the Hopf point in two parameters

4. Click **Grab**, press `Tab` three times until the bottom line shows **Lab 4 … HB**, then press `Enter`.
5. Click **Run**. A menu titled **Hopf Pt** appears; choose **Two Param**. AUTO follows the Hopf point as `phi` falls from 0.333 towards 0 while `iapp` rises to about 0.266 (new labels 6–9).

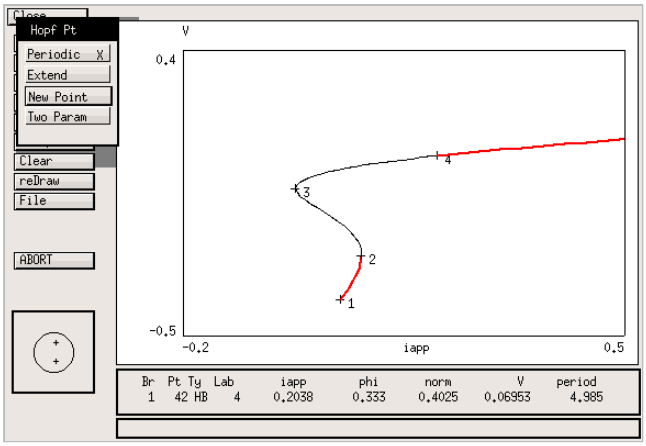


Figure 2. The menu shown for a Hopf point. Two Param continues the bifurcation in the two parameters.

6. Click **Axes** and choose **Two par** (press **Ok** if a range box appears). The diagram now shows `phi` against `iapp`: the blue line is the Hopf curve. Parameter pairs on one side of it give a stable rest state; on the other side the cell oscillates.

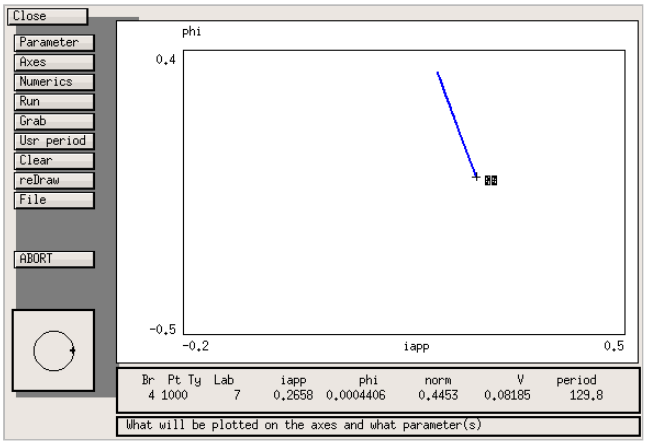


Figure 3. The Hopf curve in the (`iapp`, `phi`) plane, here after the optional restart from Lab 7 (part D).

D. Optional: restart from a point on the curve

7. Click **Grab**, press `Tab` until the bottom line shows **Lab 7**, then press `Enter`.
8. Click **Run → Two Param**. The curve is continued further (labels 10–13).

> **Avoid** *Extend* and *Periodic* from a point on the two-parameter curve (step 8): a long-standing bug in XPPAUT's AUTO can break the session. Reload the page to recover.

Tips

- **See AUTO's numbers:** Settings → Output → Side panel shows the table AUTO prints (branch, point, type, label, parameters).
- **Save the diagram:** File → SVG or Postscript in the AUTO window; the file downloads automatically. **Save session** keeps everything in one .zip.
- **Phones and tablets:** tap the menu items; `Tab`, `Enter` and `Esc` are on the ⌨ key strip. The site can be installed as an app and then works offline.

XPPAUT-Web (JKG-2026) · zetaone.github.io/XPPAUT-Web · source: github.com/ZetaOne/XPPAUT-Web (GPL v2).
XPPAUT © Bard Ermentrout; AUTO by E. J. Doedel. Morris–Lecar model: `lecar.ode` from the XPPAUT examples.
Screenshots from the XPPAUT-Web test suite.